

The Neuroscience of Relational N-Back

Brain Network Mapping and Theoretical Analysis of Gf-Boosting Potential

A research report on the GOATED Relational N-Back training application. Maps every game mode and relation type to its recruited brain networks, and evaluates which configuration has the strongest theoretical case for fluid-intelligence transfer.

TL;DR

- **Standard N-back recruits the bilateral frontoparietal "multiple-demand" network** (DLPFC BA 9/46, VLPFC BA 44/45, frontopolar BA 10, dorsal ACC BA 32, premotor BA 6, posterior parietal BA 7/40, anterior insula, medial cerebellum) plus modality-specific buffers (Owen et al. 2005; Rottschy et al. 2012; Duncan 2010). Relational N-Back's exotic modes layer additional, theoretically distinct networks on top: hippocampal–entorhinal cognitive-map circuitry for RINT, IPS for quantitative relations, ATL semantic hub for verbal relations, PPA/RSC for spatial relations, STG/Heschl's for sound, and right superior parietal/dorsolateral premotor for 3D and projected 4D Alien modes.
- **RINT (Relational Integration) is the mode with the strongest theoretical case for Gf transfer.** It is the only mode that simultaneously taxes the three substrates most consistently associated with general fluid intelligence: rostralateral PFC / frontopolar BA 10 (Christoff et al. 2001; Bunge, Helskog & Wendelken 2009), the hippocampal–entorhinal cognitive map for abstract relational structure (Constantinescu, O'Reilly & Behrens 2016; Park, Miller & Boorman 2021; Garvert, Dolan & Behrens 2017), and the Halford "ternary-relational" workload that constitutes the upper bound of human relational processing capacity (Halford, Wilson & Phillips 1998). It also maps almost directly onto the RFT/SMART training operations (more-than/less-than, transitive derivation) that have produced the largest reported IQ effects in the published literature (Cassidy, Roche & Hayes 2011; Colbert, Tyndall, Roche & Cassidy 2018) — although those effects shrink substantially under stringent controls (Roche, Cummins, Cassidy, Dillon, Moore & Grey 2023; May et al. 2022).
- **Pure N-back-to-Gf transfer is small and contested.** Au et al. (2015, meta-analysis of 20 n-back studies) reported Hedges' $g = 0.24$, which the authors framed as a small but practically meaningful effect (≈ 3.6 IQ points on a 15-SD scale, an inference from their effect size rather than a verbatim figure); Melby-Lervåg, Redick & Hulme (2016) and Redick, Shipstead, Harrison et al. (2013, *J Exp Psychol Gen* 142:359–379 — "despite improvements on both the dual n-back and visual search tasks with practice, and despite a high level of statistical power, there was no positive transfer to any of the cognitive ability tests") argue the effect collapses under active controls. The plausible mechanistic upgrade in this app is *not* greater working-memory load but the move from *maintenance/updating* to *relational integration over a derived fact graph*, which is the cognitive operation Gray, Chabris & Braver (2003), Christoff et al. (2001), and Duncan (2010) identify as the actual rate-limiter on Gf.

1. Background: What Standard N-Back Actually Trains

Owen, McMillan, Laird & Bullmore (2005), in the canonical ALE meta-analysis of 24 fMRI N-back studies, identified seven regions activated regardless of stimulus content or process: (1) bilateral and medial posterior parietal cortex (BA 7/40), (2) bilateral premotor cortex (BA 6), (3) dorsal cingulate / medial premotor cortex (BA 32/6), (4) bilateral rostral/frontopolar prefrontal cortex (BA 10), (5) bilateral dorsolateral prefrontal cortex (BA 9/46), (6) bilateral mid-ventrolateral prefrontal cortex (BA 45/47), and (7) medial cerebellum. Rottschy et al. (2012) confirmed and extended this, and Yapple, Stevens & Arsalidou (2019) showed the same pattern across the adult lifespan with age-related compensatory shifts.

This set is essentially the **multiple-demand (MD) network** (Duncan 2010, 2013; Fedorenko, Duncan & Kanwisher 2013), the same set of regions whose damage produces selective fluid-intelligence loss (Woolgar et

al. 2010; Woolgar, Duncan, Manes & Fedorenko 2018) and whose activity scales with Gf (Assem, Blank, Mineroff, Ademoglu & Fedorenko 2020).

Gray, Chabris & Braver (2003, *Nature Neuroscience* 6:316–322) showed using 3-back specifically that lateral prefrontal and parietal MD regions *mediate* the relationship between Gf (Raven's APM) and accuracy under high-interference trials — i.e., the brain machinery that distinguishes high-Gf from low-Gf individuals is the same machinery N-back loads. This is the strongest mechanistic case for *why* N-back might transfer to Gf, and the weak transfer findings (Au et al. 2015; Jaeggi et al. 2008, 2010) are consistent with this account.

What standard N-back fails to do — and what this app's exotic modes attempt — is engage **relational integration** in the Halford/Christoff/Bunge sense. Standard N-back is an *updating + recognition* task; it does not require considering multiple relations *jointly*. That distinction matters enormously for the IQ-transfer question.

2. Brain Network Mapping Tables

2a. Game Modes

Mode	Primary Networks	Key Specific Regions	Evidence / Citations
Normal N-Back	Frontoparietal MD / central executive; phonological loop or visuospatial sketchpad (modality-dependent)	DLPFC (BA 9/46), VLPFC (BA 44/45), frontopolar BA 10, dorsal ACC (BA 32), posterior parietal (BA 7/40), premotor BA 6, medial cerebellum, anterior insula	Owen et al. 2005; Rottschy et al. 2012; Yapple, Stevens & Arsalidou 2019; Gray, Chabris & Braver 2003
Type N-Back (per-category queues)	MD network + binding / source-monitoring; multiple-buffer parallel maintenance	DLPFC for set maintenance, anterior PFC / BA 10 for branching (sub-goal maintenance) , parahippocampal cortex for source tagging	Koechlin et al. 1999 (frontopolar cortex and branching); Braver & Bongiolatti 2002; Burgess et al. 2007 (lateral BA 10 = managing multiple ongoing task sets)
RINT (Relational Integration)	Apex of frontoparietal hierarchy + hippocampal–entorhinal cognitive map	Left rostrolateral PFC (BA 10p) for integrating ≥ 2 relations; bilateral hippocampus (esp. anterior) and entorhinal cortex for transitive inference and grid-like coding of abstract relations; vmPFC/mOFC for value/structure; intraparietal sulcus (IPS) for first-order relations	Christoff et al. 2001; Bunge, Helskog & Wendelken 2009; Wendelken & Bunge 2010; Waltz et al. 1999; Heckers et al. 2004; Zalesak & Heckers 2009; Zeithamova, Schlichting & Preston 2012; Constantinescu, O'Reilly & Behrens 2016; Park, Miller & Boorman 2021; Garvert, Dolan & Behrens 2017
Mixed N-Back (Normal/Type per-trial)	MD + task-switching network	DLPFC, inferior frontal junction (IFJ) for task-set reconfiguration, pre-SMA, superior parietal (rule-set update), dorsal ACC for switch-cost monitoring	Braver, Reynolds & Donaldson 2003; Kim et al. 2011; Schwarze, Fandakova & Lindenberger 2024
Mixed RINT (3-way switch)	MD + switching + relational integration	All of the above + recurrent RLPFC engagement; raises sustained "mixing cost" load	Braver et al. 2003 (sustained vs transient task-switching); Christoff et al. 2001; Bunge et al. 2009
Impossible (multi-stream, per-stream random mode)	MD + switching + binding under interference	DLPFC/dACC for conflict, anterior insula (salience network) for stream selection, hippocampus for binding-with-source, RLPFC for branching across streams	Dosenbach et al. 2007 (salience); Seeley et al. 2007; Koechlin et al. 1999; Burgess et al. 2007
Binary Logic (AND/OR/XOR/AND_NOT pairings)	RLPFC + MD + deductive-reasoning network	Left RLPFC (BA 10) specifically for higher-order combination of two relational outcomes; left IFG (BA 44/45) for logical syntax; parietal BA 7	Goel & Dolan 2001; Knauff et al. 2003; Christoff et al. 2001; Hampshire et al. 2011 (dynamic network for relational integration)
Alien Cube (3×3×3, position + relation match)	MD + visuospatial sketchpad + mental-rotation network	Right superior parietal lobule (BA 7), bilateral IPS, dorsolateral premotor (BA 6), hippocampus for spatial binding, parahippocampal place area (PPA) for cell-layout	Shepard & Metzler tradition; Hawes et al. 2019 (ALE meta of mental rotation); Suchan et al. 2006 (3D cube WM); Epstein 2008 (PPA)
Alien Square (3×3)	MD + visuospatial sketchpad	Right posterior parietal, IPS, dorsal stream (occipitoparietal), DLPFC	Awh & Jonides 2001; Smith & Jonides 1997
Alien Tesseract (projected 4D, 81 cells)	MD + bilateral parietal + premotor; presumed extension of mental-rotation circuitry to non-3D coordinates	Bilateral posterior parietal (BA 7), IPS, dorsolateral premotor; hippocampus likely recruited heavily for non-canonical spatial structure	Aflalo & Graziano 2008 (behavioral 4D path integration — no fMRI); Ambinder et al. 2009; inferred from 3D mental-rotation network
Variable N (N drifts ± 1)	MD + flexible rule-set maintenance	DLPFC for rule maintenance, RLPFC for monitoring current-N goal, ACC	Sakai & Passingham 2003; Bunge 2004 (rule representation)

Mode	Primary Networks	Key Specific Regions	Evidence / Citations
Adaptive N (auto-titrating across sessions)	MD network operating at edge of capacity; engages neuroplasticity	Striatum (esp. caudate) for skill-learning related update; DLPFC for capacity-pushing trials	Dahlin et al. 2008; Salminen, Kühn et al. 2016 (striatal change predicts dual-N-back transfer)
Distractors (near-miss interference)	MD + inhibitory control + lure-resolution	Left VLPFC (BA 45/47) for proactive interference resolution; anterior hippocampus for pattern separation; dACC for conflict	Jonides & Nee 2006; Nee & Jonides 2008; Bakker, Kirwan, Miller & Stark 2008 (CA3/DG pattern separation)

2b. Relation Types

Relation Type	Primary Networks	Specific Regions	Evidence
Spatial 2D (inside, overlapping, mirrored, surrounded)	Dorsal visual stream + visuospatial sketchpad	Right superior parietal lobule, IPS, occipitoparietal junction, right posterior parietal cortex	Kosslyn et al. 1989; Hawes et al. 2019
Spatial 3D (orbiting, gravity-bound, ascending spiral)	Dorsal stream + mental-rotation network + PPA	Right superior parietal (BA 7), bilateral IPS, dorsolateral premotor (BA 6), PPA, retrosplenial cortex for ego/alto transformations	Zacks 2008; Epstein 2008; Suchan et al. 2006
Trait (same color, hollow/solid, size, stripes, shadow copy)	Ventral visual stream + RLPFC for cross-trial trait binding	Fusiform (color: V4/V8), lateral occipital complex (shape), inferior temporal cortex; for "same/different" abstraction → RLPFC	Grill-Spector & Weiner 2014; Pollmann 2004; Christoff et al. 2001 (relational comparison)
Quantitative (2:1, pyramid, balanced scale, increasing row)	IPS + MD network	Horizontal segment of the intraparietal sulcus (hIPS) for approximate number; bilateral IPS for proportions; DLPFC for relational comparison	Dehaene, Piazza, Pinel & Cohen 2003; Piazza, Izard, Pinel, Le Bihan & Dehaene 2004; Pinel et al. 2001; Hawes et al. 2019
Verbal — Semantic (causes, part of, negates)	Anterior temporal lobe semantic hub + language network	Bilateral ventral ATL (semantic hub) ; left IFG (BA 45/47) for semantic control; left middle temporal gyrus	Patterson, Nestor & Rogers 2007; Lambon Ralph, Jefferies, Patterson & Rogers 2017; Jefferies 2013; Binder et al. 2009
Verbal — Comparison (bigger, hotter, older)	RLPFC + IPS (magnitude comparison)	Left RLPFC for cross-trial relation comparison; bilateral IPS for ordinal/magnitude representation	Wendelken, Nakhachenko, Donohue, Carter & Bunge 2008; Fias et al. 2003; Knops et al. 2009
Verbal — Temporal (before, follows, precedes)	Hippocampus (sequence memory) + IPS + RLPFC	Hippocampus and entorhinal for temporal-order coding; right IPS for serial-order; RLPFC for relational evaluation	MacDonald et al. 2011; Hsieh et al. 2014; Ekstrom & Ranganath 2018
Verbal — Directional (north of, NW of, inside of)	Retrosplenial + parahippocampal + parietal	RSC for allocentric reference frame, PPA for layout, IPS for directional coding, dorsolateral PFC for rule maintenance	Epstein 2008; Vass & Epstein 2013; Marchette et al. 2014
Sound (pitch, rhythm)	Auditory cortex + cerebellum	Anterolateral Heschl's gyrus + STG for pitch; planum temporale + cerebellum + SMA for rhythm/timing	Patterson, Uppenkamp, Johnsrude & Griffiths 2002; De Angelis et al. 2018; Grahn & Brett 2007
Heterogeneous token types (real words, nonsense, emoji, unicode, alphanumeric, voronoi)	Forces relational system away from semantic crutches; pushes load onto domain-general MD/RLPFC machinery	When semantic content is degraded, ATL recruitment falls and frontoparietal/RLPFC recruitment rises; same logic Christoff et al. (2001) used with abstract figural matrices	Christoff et al. 2001; Krawczyk 2012 (review of relational reasoning networks); Cassidy/SMART rationale

3. Theoretical Analysis: Which Mode Has the Highest Gf Potential?

3.1 The Empirical Landscape

Two large meta-analyses anchor the debate. Au et al. (2015, *Psychon Bull Rev* 22:366–377) found a small but reliable Gf transfer effect from n-back training (Hedges' $g \approx 0.24$, which on a 15-point IQ-SD scale implies roughly 3–4 IQ points of equivalent gain — an inference from the effect size rather than a verbatim claim by the authors, who framed it as “since Gf is a fundamental cognitive skill that underlies a wide range of life functions, even small improvements can have profound societal ramifications”). Melby-Lervåg & Hulme (2013), Melby-Lervåg, Redick & Hulme (2016, *Perspect Psychol Sci* 11:512–534), Redick, Shipstead, Harrison et al. (2013, *J Exp Psychol Gen* 142:359–379), and Dougherty, Hamovitz & Tidwell (2016) argue that once active control groups and baseline-corrected effect sizes are used, the far-transfer effect to nonverbal ability is not reliably distinguishable from zero. Redick et al. specifically: “Despite improvements on both the dual n-back and visual search tasks with practice, and despite a high level of statistical power, there was no positive transfer to any of the cognitive ability tests” — across 20 sessions of adaptive dual n-back versus active visual-search placebo. The honest read: standard dual N-back probably produces *some* transfer to fluid reasoning under some conditions, but the effect is small, fragile, and heavily moderated by control type, expectancy effects, motivation, and individual differences (Jaeggi et al. 2014).

Crucially, the cause of the weak transfer is *not* mysterious. Standard N-back loads working-memory **updating and recognition**, but Gf as measured by Raven's APM is fundamentally a test of **relational integration** — joint consideration of multiple relations to derive a conclusion (Halford, Wilson & Phillips 1998; Carpenter, Just & Shell 1990). Working-memory capacity and Gf share variance, but the shared variance lives primarily in the relational-integration / interference-control component (Gray, Chabris & Braver 2003; Burgess et al. 2011), not in the maintenance buffer.

3.2 The Mechanism that Matters

There is now strong convergent evidence — patient (Waltz et al. 1999; Krawczyk et al. 2008), neuroimaging (Christoff et al. 2001; Kroger et al. 2002; Bunge, Helskog & Wendelken 2009; Wendelken et al. 2008; Hampshire et al. 2011), developmental (Crone et al. 2009; Wendelken et al. 2011), and theoretical (Halford et al. 1998; Duncan 2010, 2013) — that **the rostrolateral prefrontal cortex / frontopolar BA 10 is the apex of a hierarchy that integrates outputs of lower-order relational operations**. Left RLPFC specifically scales with the relational-integration demand (Bunge et al. 2009), not with overall task difficulty. Ternary-relational items (where three variables must be jointly considered) are the workload that defines adult capacity (Halford, Wilson & Phillips 1998; Andrews & Halford 2002), and they are precisely what Raven's matrices and analogy problems require.

A parallel and complementary story comes from the medial temporal lobe. Heckers, Zalesak, Weiss, Ditman & Titone (2004), Zalesak & Heckers (2009) and Zeithamova, Schlichting & Preston (2012) show hippocampal engagement during transitive inference. Constantinescu, O'Reilly & Behrens (2016, *Science* 352:1464–1468) demonstrated that the entorhinal cortex uses **hexagonal grid-like codes to represent abstract two-dimensional conceptual spaces**, with the same hexadirectional fMRI signature seen during spatial navigation — “Humans navigating conceptual two-dimensional knowledge showed the same hexagonal signal in a set of brain regions markedly similar to those activated during spatial navigation.” Bellmund, Gärdenfors, Moser & Doeller (2018, *Science* 362:eaat6766) extended this into a general theory of “cognitive spaces”. Park, Miller & Boorman (2021, *Nature Neuroscience* 24:1292–1301) showed that **discrete, abstract social hierarchies with transitive ordering are represented as a unitary cognitive map in hippocampus and entorhinal cortex, with grid-like codes in entorhinal cortex and medial PFC supporting novel inferred decisions**. Garvert, Dolan & Behrens (2017, *eLife* 6:e17086) demonstrated that the hippocampal–entorhinal system extracts statistical regularities from discrete relational structures and supports transitive inference.

These findings matter because the **RINT mode is essentially a real-time transitive-inference task layered onto working-memory load**. Each trial requires (a) maintaining an evolving fact graph over named entities (α , β , γ ...) — a hippocampal–entorhinal cognitive-map operation; (b) integrating the current premise with N prior premises to derive a transitive conclusion — a left RLPFC operation; (c) gating that result against working-memory contents — a DLPFC/IPS operation. **No other mode in the app — and indeed no standard cognitive-training task — combines those three demands simultaneously.**

3.3 The RFT/SMART Convergence and Its Caveats

Relational Frame Theory (Hayes, Barnes-Holmes & Roche 2001) explicitly identifies derived relational responding — including transitive frames of comparison (more-than/less-than) — as the core operant of higher cognition. SMART training (Cassidy, Roche & Hayes 2011; Cassidy, Roche, Colbert, Stewart & Grey 2016) has produced some of the largest reported IQ effects of any cognitive training paradigm. Colbert, Tyndall, Roche & Cassidy (2018, *J Behav Educ* 27:509–531) reported “a mean WISC-IV Full Scale IQ increase of 23 points in a sample of 15 school children aged 11–12 years, with the smallest increase for any participant being 14 points (i.e., approx. one standard deviation)”; the same study reported significant WASI Verbal IQ gains (alongside Matrix Reasoning and Vocabulary) for the experimental group only (n=13), with no significant change in controls. Hayes & Stewart (2016, *Br J Educ Psychol* 86:397–411) reported significant gains under an active control (Scratch coding) at n=28. McLoughlin, Tyndall & Pereira (2021) reported nonverbal IQ gains in a stratified active-controlled trial.

But the evidence must be reported honestly. May, Tyndall, McTiernan, Roderique-Davies & McLoughlin (2022, *Br J Educ Technol* 53:1244–1261) meta-analyzed five SMART trials and reported “a moderate impact of SMART on measures of nonverbal IQ ($g = 0.57$, 95% CI [0.24, 0.89])” — but rated **all** five studies at high risk of bias. Most importantly, **Roche, Cummins, Cassidy, Dillon, Moore & Grey (2023, *J Contextual Behav Sci* 28:185–197)**, by the program's own developers, reported: “The SMART effect was replicated, but not after controlling for baseline intelligence or attentional skills.” That paper is the strongest available evidence that previously-claimed large SMART IQ gains substantially reflect baseline imbalances and individual differences in attention rather than a true intelligence transfer.

The correct interpretation: relational training is the most theoretically promising route to IQ transfer, the operations it trains are the operations Gf instruments measure, and reported effect sizes are larger than for standard WM training — but the empirical base is thin, biased, and partially deflates under controls. The RINT mode is best understood as a high-load, RFT-style relational training task with an additional working-memory loading factor; it shares the theoretical strengths and the empirical caveats.

3.4 Why Not Just Load Working Memory Harder?

Higher N produces larger MD/DLPFC activations (Owen et al. 2005; Yaple et al. 2019) but yields diminishing returns on Gf transfer (Jaeggi et al. 2014; Au et al. 2015). The Salminen, Kühn et al. (2016, *J Neurosci*) finding that **striatal activation change** predicts dual N-back transfer, not the magnitude of DLPFC change, suggests that procedural automatization, not maximal cortical load, mediates the transferable component. Pure load saturates a system; it does not necessarily change what the system can integrate.

The relational-complexity argument (Halford, Wilson & Phillips 1998) predicts the opposite of pure load: increases in *the number of variables jointly bound* should produce the largest gains in the limit on abstract reasoning, because that is exactly the dimension Gf measures.

3.5 The Argument for Mode Switching

Mixed, Mixed RINT, and Impossible modes recruit the **task-switching network** (Braver, Reynolds & Donaldson 2003; Kim et al. 2011) on top of all the demands of their constituent modes. Cognitive-flexibility training has its own modest IQ-transfer literature: Karbach & Verhaeghen (2014, *Psychological Science* 25:2027–2037), a meta-analysis of 49 executive-function/WM training studies in older adults (>60 years), reported far-transfer effects significant in 3 of 4 comparisons with mean standardized differences of MSD =

0.2–0.3; Karbach & Kray (2009, *Dev Sci* 12:978–990) found “substantial far transfer to other executive tasks and fluid intelligence in all age groups” from task-switching training, though later replications recovered near but not far transfer. However, *task-switching alone* does not produce robust Gf transfer in active-controlled meta-analyses (Melby-Lervåg, Redick & Hulme 2016). The most plausible role of switching demands here is not as a primary Gf driver but as a forcing function that prevents subjects from chunking or routinizing the task — thereby preserving the relational-integration demand at full intensity rather than letting it decay into pattern recognition.

3.6 The Argument for Stimulus Heterogeneity

The app's 7 token types (real words, pronounceable nonsense, emoji, unicode, alphanumeric strings, voronoi patterns) and 6 relation categories implement what RFT calls “multiple exemplar training” and what cognitive psychology calls **abstraction-forcing variability** (Schmidt & Bjork 1992; Cassidy, Roche & Hayes 2011). When the surface stimuli vary unpredictably, performance cannot bootstrap from modality-specific representations; it must use the domain-general relational operation. Neurally, this should suppress ATL/PPA/STG modality-specific activation and concentrate processing in MD + RLPFC — exactly the regions Gray, Chabris & Braver (2003) and Duncan (2010) identify as Gf-supporting.

3.7 The Argument Against Alien Tesseract as the Top Mode

Alien Tesseract is intellectually impressive but its plausible Gf payoff is bounded. Aflalo & Graziano (2008, *J Exp Psychol HPP* 34:1066–1077) and Ambinder, Wang, Crowell, Francis & Brinkmann (2009, *Psychon Bull Rev* 16:818–823) showed humans can learn 4D path-integration and angle judgments behaviorally — “subjects were able to learn the spatial geometry of the 4-dimensional maze as measured by their ability to perform path integration” (Aflalo & Graziano 2008) — but there is **no published fMRI study of a distinct 4D neural representation**; the inferred substrate is generalized parietal/premotor mental-rotation machinery (Hawes et al. 2019). High novelty drives effort and engagement, but the 4D demand is largely an extra visuospatial-binding load rather than an additional relational-integration demand in Halford's sense. It will likely train spatial working memory and right parietal cortex more than it will train Gf.

3.8 The Synthesis Recommendation

The mode-and-configuration combination with the strongest theoretical case for Gf transfer is:

Mixed RINT with high relational-complexity content (verbal comparison + quantitative + spatial-3D relations mixed), heterogeneous token types, adaptive-N titration, and distractors enabled.

This configuration:

1. Forces ternary-relational integration on each trial (the Halford ceiling of human capacity).
2. Maximally recruits left RLPFC (Bunge et al. 2009) and the hippocampal–entorhinal cognitive map (Park et al. 2021; Constantinescu et al. 2016).
3. Maintains MD-network load via adaptive N and the working-memory window over the fact graph (Gray, Chabris & Braver 2003).
4. Suppresses modality-specific shortcuts through token heterogeneity, forcing the operations into the domain-general relational machinery.
5. Adds task-switching demands (Braver et al. 2003) that prevent automatization without dominating the load.
6. Uses interference (distractors) to engage hippocampal pattern separation and VLPFC interference resolution — the components Burgess et al. (2011) and Gray et al. (2003) identify as mediating the WM–Gf overlap.

Pure single-mode RINT is the second-best option and is the cleanest training stimulus for the relational system specifically; Mixed RINT trades a small loss in dose-on-target for substantial gains in transfer breadth and resistance to strategy-narrowing.

4. Recommendations (Concrete, Staged)

Stage	Sessions	Configuration	Threshold to advance
Stage 1 — Calibration	1–5	Single-mode RINT at N=2 with all six relation categories rotating, mixed token types, distractors <i>off</i> . Verify reliable basic transitive-derivation operation ($>75\%$ d').	80% accuracy at N=2 across two consecutive sessions.
Stage 2 — Load	6–20	Adaptive RINT, N titrating between 2 and 4, distractors on, all relation types.	Sustained $N \geq 3$ with $>70\%$ accuracy.
Stage 3 — Transfer-oriented	21+	Mixed RINT (Normal/Type/RINT switching) with adaptive N, heterogeneous tokens, full relation set, distractors on. The configuration with the strongest theoretical case.	Ongoing; reassess every 10 sessions with unseen-form Gf measure.

Benchmarks that would change the protocol

- If accuracy plateaus below 60% on RINT N=2 for three consecutive sessions, drop to single-relation-type RINT to scaffold the relational operation before re-mixing.
- If pre/post Raven's APM (or equivalent unseen-form Gf measure) at the 20-session mark shows zero gain *and* the within-task adaptive N has plateaued, the training is failing to produce both near and far transfer; switch to Binary Logic mode, which has the strongest theoretical case for an alternative driver (deductive-reasoning network engagement).
- Use Alien Cube/Tesseract modes sparingly ($\leq 20\%$ of training time) — they have strong novelty/engagement value but the smallest theoretical Gf payoff per minute.

5. Caveats

1. The Gf-transfer claim is contested even at its source. Au et al. (2015, $g = 0.24$) implied a ~3–4 IQ-point gain from standard N-back, but Melby-Lervåg, Redick & Hulme (2016), Redick et al. (2013) and Dougherty et al. (2016) argue this collapses under active-control comparison and baseline correction. Any IQ-boosting claim for this app, however theoretically motivated, is a prediction not an established fact.

2. The largest reported relational-training IQ effects (SMART, Cassidy et al. 2011; Colbert et al. 2018) come from underpowered, mostly passive-control studies rated as high-bias by the only meta-analysis (May et al. 2022). Roche et al. (2023), by the SMART developers themselves, found the IQ gain disappears when baseline IQ and attention are controlled. The honest expected effect size from a SMART-like relational protocol is closer to small than to a standard deviation.

3. The brain-network mapping is from contrast studies of task performance, not training. Saying mode X recruits region Y does not automatically mean training mode X strengthens region Y or transfers from it. The training-induced plasticity literature (Klingberg 2010; Constantinidis & Klingberg 2016) shows fronto-parietal change but inconsistent transfer.

4. There is no published fMRI study of 4D / tesseract cognition. Aflalo & Graziano (2008) and Ambinder et al. (2009) showed behavioral learning of 4D structure but the neural substrate is inferred from the 3D mental-rotation literature.

5. Many of the relation categories (e.g., “shadow copy”, “ascending spiral”, “voronoi patterns”) have no published fMRI study using exactly that operation. The mapping in the tables uses the closest validated analog (mental rotation, color discrimination, spatial binding, etc.). Treat the regional assignments as *theoretically motivated predictions*, not measured findings.

6. Striatal involvement and motivation matter. Salminen, Kühn et al. (2016) found that *striatal* change predicted dual-N-back transfer, not cortical change. Dopaminergic / motivational engagement may be at least as important as the cortical engagement story above, and adaptive-difficulty mechanisms in the app are

doing real mechanistic work on that channel.

7. Individual differences dominate. Jaeggi et al. (2014) and Au et al. (2020) show that motivation, implicit theories of intelligence, baseline ability, and conscientiousness predict who benefits. No mode is universally optimal; the recommendations above are central-tendency predictions for typical adult subjects.